

Technical Description (Draft)

Governing Equations

We model the DC railway traction network as a resistive network with current sources and nonlinear loads.

The core equation is the nodal form of **Kirchhoff's current law (KCL)**:

$$Y \cdot V = J$$

where:

- Y is the nodal admittance matrix,
 - V is the vector of node voltages,
 - J is the current injection vector.
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Substation Model

A substation is modeled as an ideal EMF E in series with an internal resistance R .

This can be written as a Norton equivalent:

$$I = G \cdot (E - V)$$

where:

- $G = \frac{1}{R}$ is the conductance,
- V is the node voltage at the feeding point.

The corresponding **power balance** is:

- Output power to the network:

$$= V \cdot I$$

- Internal loss:

$$= (E - V) \cdot I = \frac{I^2}{G}$$

Train Load Model

A train requests a power at the catenary voltage V .
The corresponding current is computed as:

$$I = \frac{P}{V}$$

At low voltages this naturally increases the current demand, but a nonlinear saturation or protection may be added:

$$I = \left(\frac{P}{V} \right)^{\alpha}$$

Iterative Solver

1. Initialize voltages with ground fixed: $V = 0$.
2. At each iteration:
 - Clear Y and J .
 - Stamp all devices (substations, lines, trains).
 - Enforce ground constraint.
 - Solve $Y \cdot V = J$.
3. Check convergence:

$$\Delta V = V^{(1)} - V^{(0)}$$

Stop if $\Delta V < \epsilon$.

Power Balance Check

At each timestep, we check:

$$P_{\text{gen}} - P_{\text{load}}$$

Any imbalance indicates a modeling or numerical error.
